SOFTWARE ENGINEERING LAB
Experiment 2 — Selecting a Software Process Model for Shield AI

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Aim: To compare software process models and select a suitable approach for Shield AI, a scam-detection system.

Process Model Comparison

The process models are compared using requirement stability, early delivery, customer feedback, technical risk and suitability for iterative improvement.

Case-Wise Analysis

CASE 1: Shield AI Core Text Analysis

The initial requirements for text submission, preprocessing and displaying an assessment are defined and can be documented before implementation.

Project Characteristics

Requirements: The core workflow is relatively stable — submit text, analyse it and display a result.

Risk: Moderate — the basic workflow is clear, but detection quality must be validated.

Customer feedback: Useful, while the basic workflow can be documented before implementation.

Selected Process Model: Waterfall Model (for a narrowly scoped baseline)

Reasons for Selection

The baseline input-validation and result-display requirements can be specified and reviewed before implementation.

A sequential baseline makes it easier to trace requirements into design and tests; detection logic still needs evaluation.

Waterfall is suitable for this stable baseline, but it is not the best single model for the full evolving scam-detection product.

CASE 2: Shield AI Detection-Rule Improvement

Shield AI detection rules and explanations may need repeated updates after reviewing sample scam and legitimate messages.

Project Characteristics

Requirements: Evolving — new scam patterns and user feedback may change priorities.

Early working software: Valuable — small improvements can be released and reviewed incrementally.

Customer feedback: Important for clarity, usability and missed or incorrect warnings.

Selected Process Model: Agile Model

Reasons for Selection

Short iterations allow the team to add and refine detection rules in manageable increments.

Feedback from test messages and users can be prioritised in the next iteration.

Agile accommodates changing scam patterns and enables repeated regression testing after each update.

CASE 3: Shield AI Privacy, Security & Detection Validation

Shield AI may process sensitive message content and produce safety-related warnings. Privacy, false positives, false negatives and external-service failures need careful risk analysis.

Project Characteristics

Risk: High — mishandling submitted content or presenting unreliable results can harm user trust and privacy.

System: Combines text analysis, optional URL lookups, safe handling of input and understandable result explanations.

Risk analysis: Required for privacy, external-service availability, false alarms, missed scams and unsafe user interpretation.

Selected Process Model: Spiral Model for high-risk validation activities

Reasons for Selection

Spiral cycles make risks explicit and require validation before expanding higher-risk capabilities.

Each cycle can evaluate privacy handling, detection errors, URL lookup failures and mitigation strategies.

This approach is useful for risk-heavy parts of Shield AI, although it may be more process overhead than needed for every small feature.

CASE 4: Shield AI User Interface Prototype

Potential users may not know which risk labels, explanations and warnings are easiest to understand until they try sample screens.

Project Characteristics

Requirements: Some interface requirements are uncertain until users view a working mock-up.

Users need to preview the text input, verdict, risk score and explanation layout before interface details are finalised.

Selected Process Model: Prototype Model

Reasons for Selection

A quick prototype lets users review sample scan and result screens before detailed UI development.

Feedback helps refine wording, warning prominence, accessibility and result explanations.

It reduces the risk of building a confusing or misleading safety interface by surfacing misunderstandings early.

Conclusion / Result

Shield AI has both a defined core workflow and evolving detection and usability needs, so no single process model is ideal for every activity. Waterfall can help document a stable baseline; Agile is suitable for improving detection rules and features in short iterations; Spiral-style risk analysis is useful for privacy, security and detection-quality risks; and prototyping is useful for validating the result interface. For the overall Shield AI project, an Agile approach with explicit risk reviews and early interface prototypes is the most suitable choice because scam patterns evolve and feedback is important. The process model should be selected based on requirement stability, delivery needs, feedback and risk.